

Saber Ganjisaffar

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Education

Ph.D. Candidate in Computer Science

University of California, Riverside (UCR)

California, US

2022 - 2027 (Expected)

M.S. in Computer Systems Architecture

Shahid Beheshti University

Tehran, Iran

2018 - 2021

Publications

SpecASan: Mitigating Transient Execution Attacks Using Speculative Address Sanitization.

ISCA 2025

S. Ganjisaffar, E. Mohammadian Koruyeh, J. Zellmer, H. Asghari-Esfeden, C. Song, N. Abu-Ghazaleh.

🏆 *IEEE Micro Top Picks 2026 (Honorable Mention)*

PRACTical: Subarray-Level Counter Update and Bank-Level Recovery Isolation for Efficient PRAC Rowhammer Mitigation.

ARXIV 2025

R. Nazaraliyev, S. Ganjisaffar, N. Nazaraliyev, N. Abu-Ghazaleh.

ShadowScope: GPU Monitoring and Validation via Composable Side Channel Signals.

ARXIV 2025

G. Almusaddar, Y. Zhang, S. Ganjisaffar, B. Williams, Y.D. Liu, D. Ponomare, and N. Abu-Ghazaleh.

Technical Skills

Languages and Frameworks: C/C++, Python, Golang, ARM/X86 Assembly, CUDA, OpenCL, Verilog, Chisel

Tools and Technologies: Linux, Bash, Git/Github, Docker, Latex, LLVM, DynamoRIO, Xilinx Vivado, ModelSim

Architectural Simulators: Gem5, GPGPU-Sim, MGPUSim, Scarab, DRAMSim3, Ramulator, CACTI

Projects

AI-Driven Performance Analysis Framework

2025-Present

A Framework for Cross-Layer Performance Analysis, Bottleneck Root Cause Discovery and Optimization.

- Developed an AI-driven automated framework to identify CPU performance bottleneck root causes by analyzing fine-grained microarchitectural event traces and generate actionable performance optimization insights for both software and hardware.
- Developed an asynchronous and efficient event tracer that captures fine-grained microarchitectural events from architectural simulators.
- Developed a modular analysis query engine and a python API that allows to run complex analysis queries on the event trace files in addition to having reusable analysis modules for automatic root cause discovery.
- Developed an Agentic AI component to automatically analyze the performance bottlenecks and reason about the underlying root causes and providing actionable insights for design space exploration loops to achieve efficient architectural designs.

CPU Architectural Modeling

2023-Present

Extending Gem5 Simulator with ARM Memory Tagging Extension (MTE) Support

- Extended the gem5 simulator with comprehensive support for the ARM MTE to create a robust platform for memory safety research.
- Modified gem5's ARM O3 CPU model, caches, and memory controller to natively handle MTE instructions and tag-checking logic.
- Enabling accurate simulation of MTE-protected software that uses LLVM toolchains like MemTagSanitizer.

GPU Architecture Modeling

2025

Design of Per-SM Hardware Performance Monitoring Units (PMUs) and On-Chip Kernel Validation for GPUs

- Designed a novel hardware architecture for low-latency, on-chip GPU kernel validation, featuring per-SM PMUs and a central Validator module.
- Modeled and evaluated the system using GPGPU-Sim for performance, CACTI for overhead, and a synthesized Verilog implementation.

Services

Artifact Evaluation Committee Member

IEEE S&P 2026, EuroSys 2026, MICRO 2025

5x Peer Reviewer

IEEE ACCESS and Elsevier Journals

Conference Organizing Team (Student Volunteer)

HPCA 2027, ASPLOS 2024

Awards

Dean's Distinguished Fellowship Award.

Awarded by University of California, Riverside (UCR) for Academic year 2022

March 2022